B. Everything Everywhere

time limit per test 1.5 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 900 | Tags: greedy, math, number theory | https://codeforces.com/problemset/problem/2226/B

An array is called *good* if the difference between the maximum value and the minimum value in the array is equal to the greatest common divisor (GCD) of all the elements in the array. Note that an empty array is considered to be not good.

More formally, an array $[a_1, a_2, \ldots, a_m]$ is *good* if and only if $\max(a_1, a_2, \ldots, a_m) - \min(a_1, a_2, \ldots, a_m) = \gcd(a_1, a_2, \ldots, a_m)$.

You are given a permutation* p of length n. Determine the number of good subarrays† in the given permutation.

*A permutation of length m is an array consisting of m distinct integers from 1 to m in arbitrary order. For example, [2,3,1,5,4] is a permutation, but [1,2,2] is not a permutation (2 appears twice in the array), and [1,3,4] is also not a permutation (m=3 but there is 4 in the array).

†An array b is a subarray of an array a if b can be obtained from a by the deletion of several (possibly, zero or all) elements from the beginning and several (possibly, zero or all) elements from the end. In particular, an array is a subarray of itself.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each testcase contains a single integer n ($2 \le n \le 2 \cdot 10^5$) — the length of the permutation p.

The second line of each testcase contains n integers $p_1, p_2, \ldots, p_n$ ($1 \le p_i \le n$) — the permutation p. It is guaranteed that p is a permutation.

It is guaranteed that the sum of n over all the test cases does not exceed $2 \cdot 10^5$.

Output

For each testcase, print a single integer — the number of good subarrays in the given permutation.

Example

Input
3 2 1 2 9 6 1 5 9 4 7 2 8 3 4 1 2 3 4

Output
1 0 3

Note

For the first testcase, only one subarray is good, which is [1, 2].

For the second testcase, it can be proven that no good subarrays exist in the given permutation.